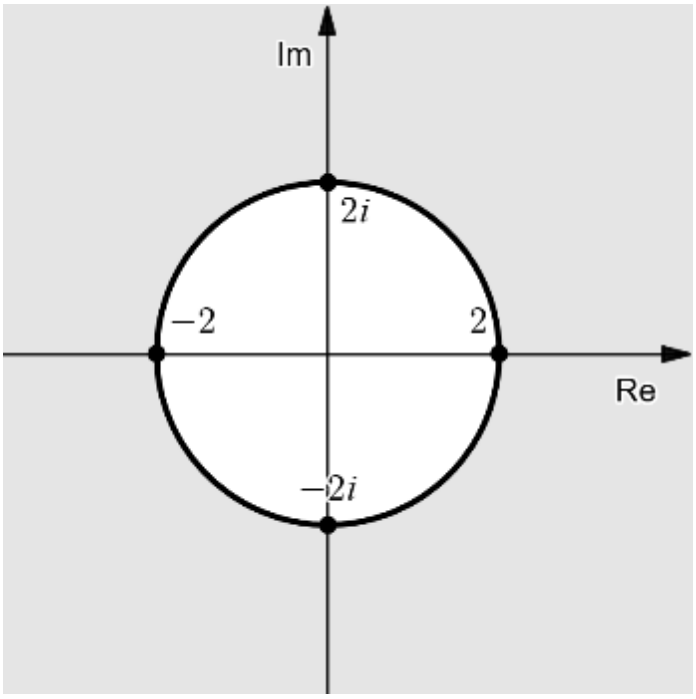
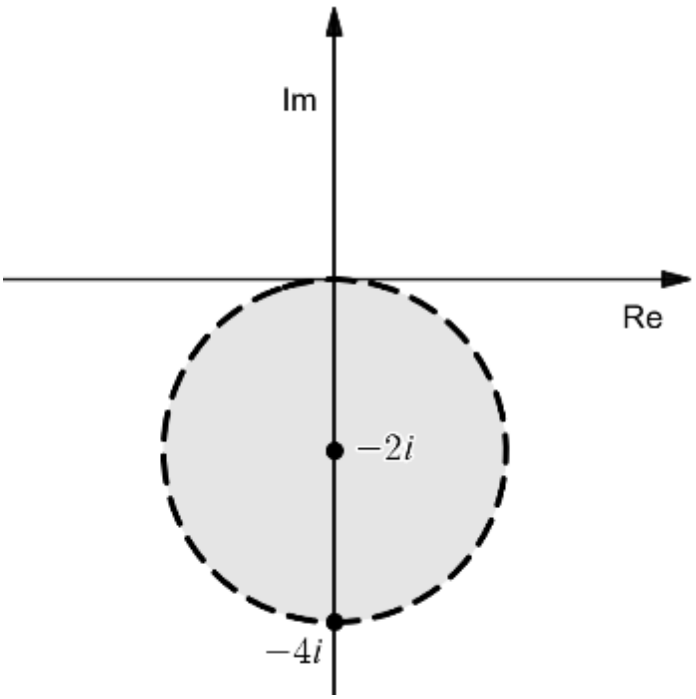
1.08 Complex Numbers: Loci - Inequalities

Worked Solutions

1 a)



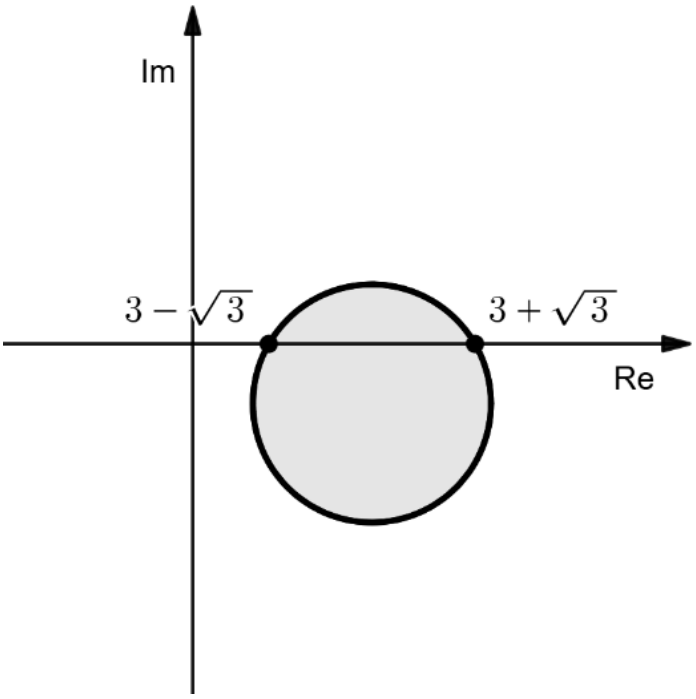
1 b)



1 c)

$$(x-3)^2+(y+1)^2=4$$

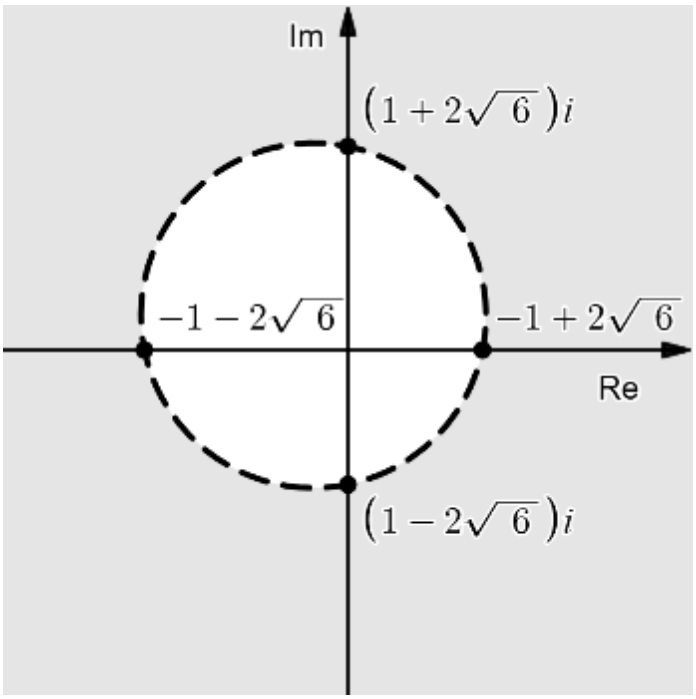
When $y=0$, $x=3\pm\sqrt{3}$



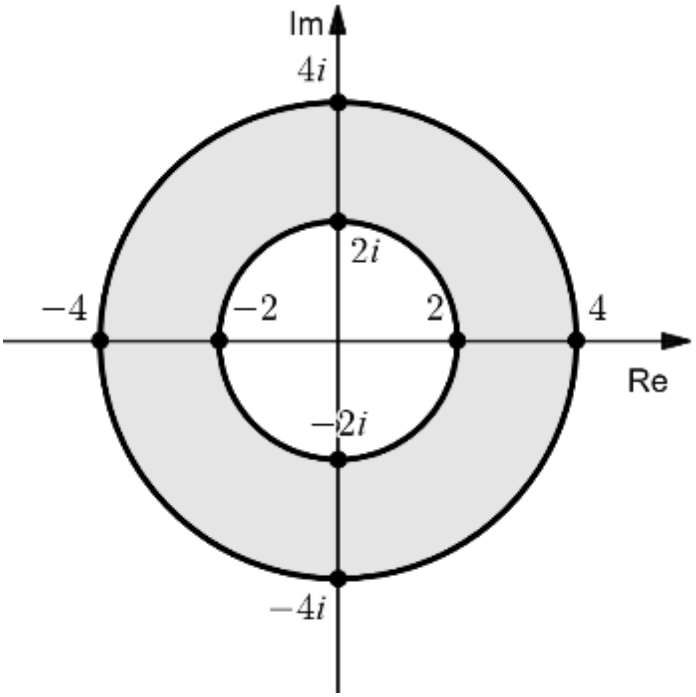
1 d)

$$(x+1)^2+(y-1)^2=25$$

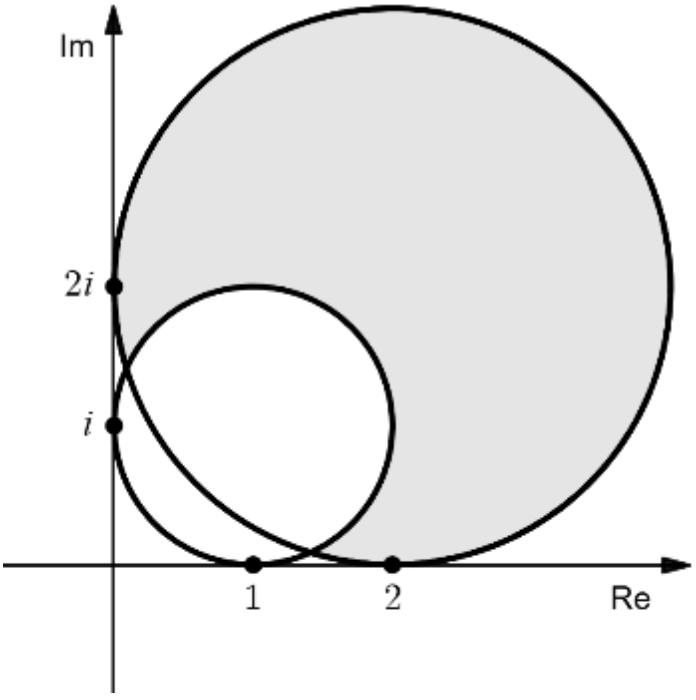
When $x=0$, $y=1\pm2\sqrt{6}$ and when $y=0$, $x=-1\pm2\sqrt{6}$



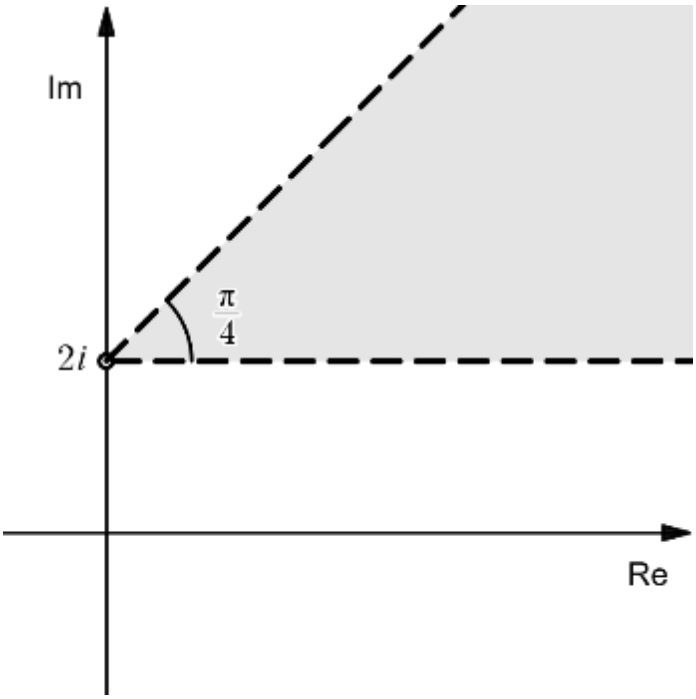
1 e)



1 f)



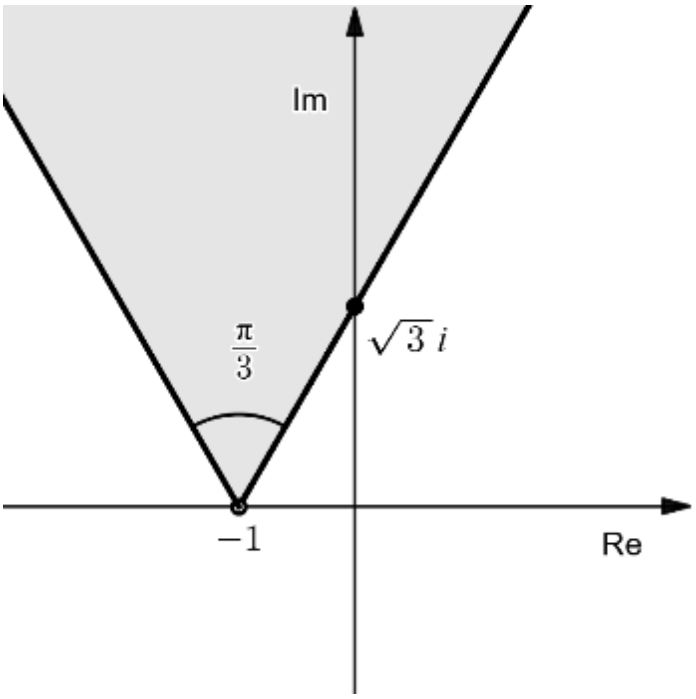
2 a)



2 b)

$$y = \tan\frac{\pi}{3}(x+1) \Rightarrow y = \sqrt{3}x + \sqrt{3}$$

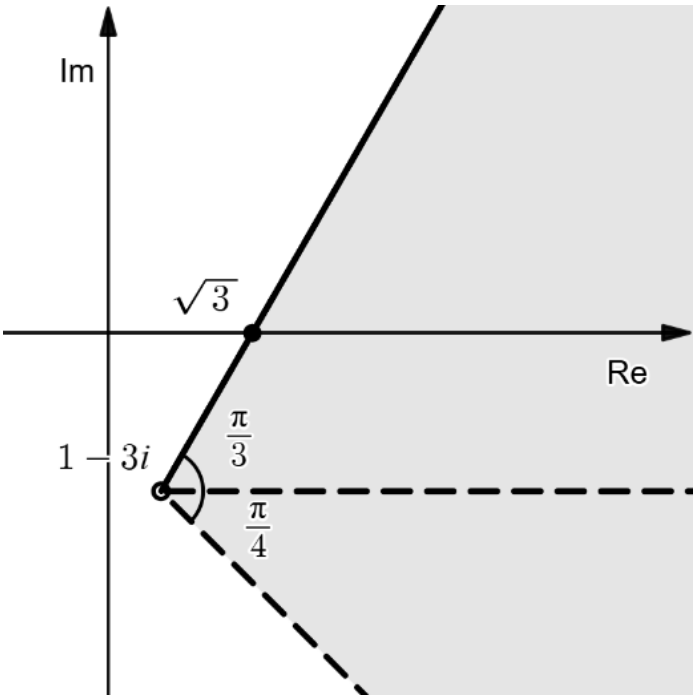
When $x = 0$, $y = \sqrt{3}$



2 c)

$$y + 3 = \tan\frac{\pi}{3}(x-1) \Rightarrow y = \sqrt{3}x - \sqrt{3} - 3$$

When $y = 0$, $x = \frac{\sqrt{3}+3}{\sqrt{3}} = \sqrt{3}$



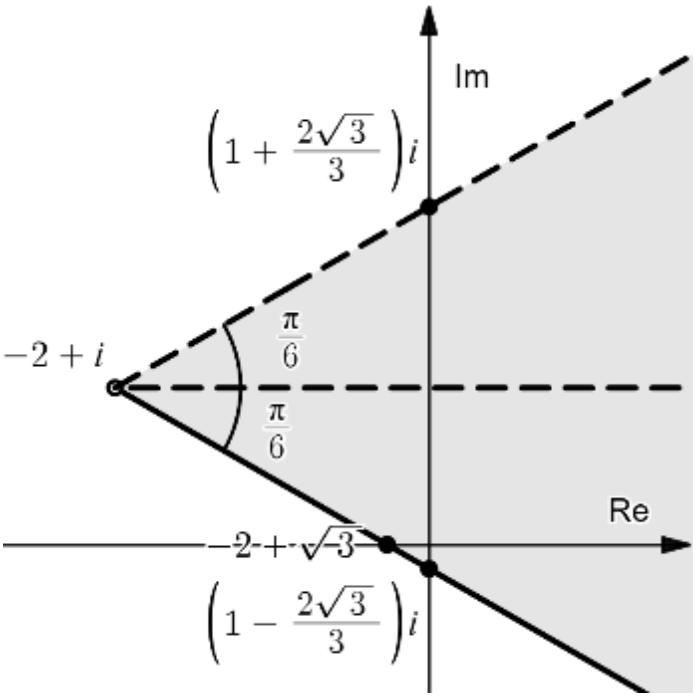
2 d)

$$y-1=\tan\frac{\pi}{6}(x+2)\Rightarrow y=\frac{\sqrt{3}}{3}x+\frac{2\sqrt{3}}{3}+1$$

When $x=0$, $y=\frac{2\sqrt{3}}{3}+1$

$$y-1=\tan\left(-\frac{\pi}{6}\right)(x+2)\Rightarrow y=-\frac{\sqrt{3}}{3}x-\frac{2\sqrt{3}}{3}+1$$

When $x=0$, $y=-\frac{2\sqrt{3}}{3}+1$ and when $y=0$, $x=\frac{-\frac{2\sqrt{3}}{3}+1}{\frac{\sqrt{3}}{3}}=2-\sqrt{3}$



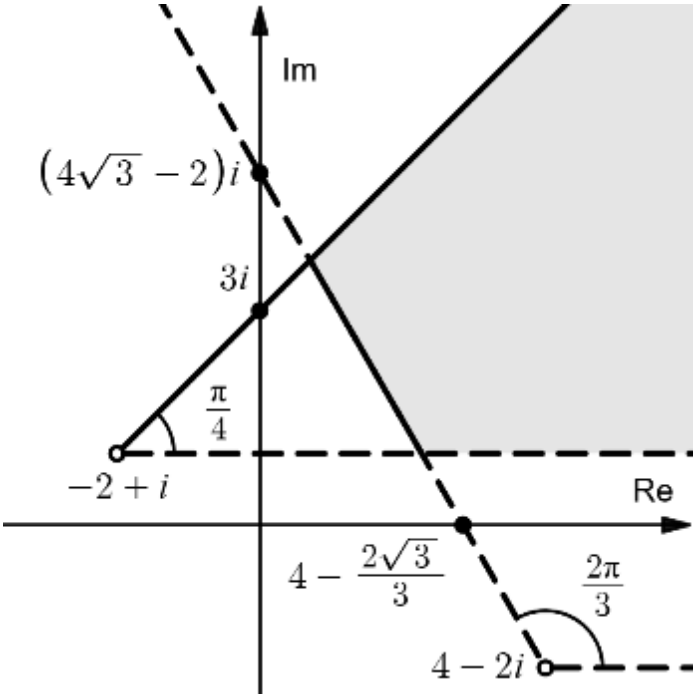
2 e)

$$y - 1 = \tan\frac{\pi}{4}(x + 2) \Rightarrow y = x + 3$$

When $x = 0$, $y = 3$

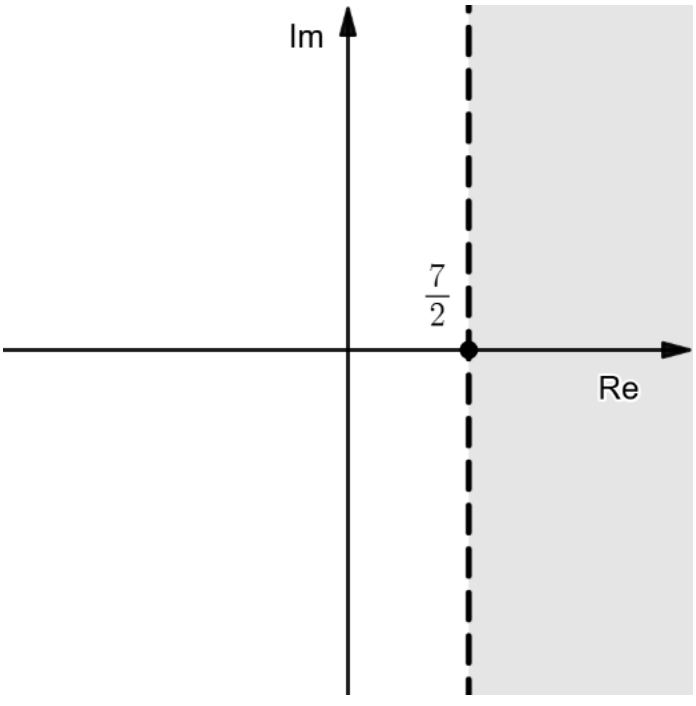
$$y + 2 = \tan\left(\frac{2\pi}{3}\right)(x - 4) \Rightarrow y = -\sqrt{3}x + 4\sqrt{3} - 2$$

When $x = 0$, $y = -2$ and when $y = 0$, $x = \frac{-4\sqrt{3}+2}{-\sqrt{3}} = 4 - \frac{\sqrt{3}}{2}$



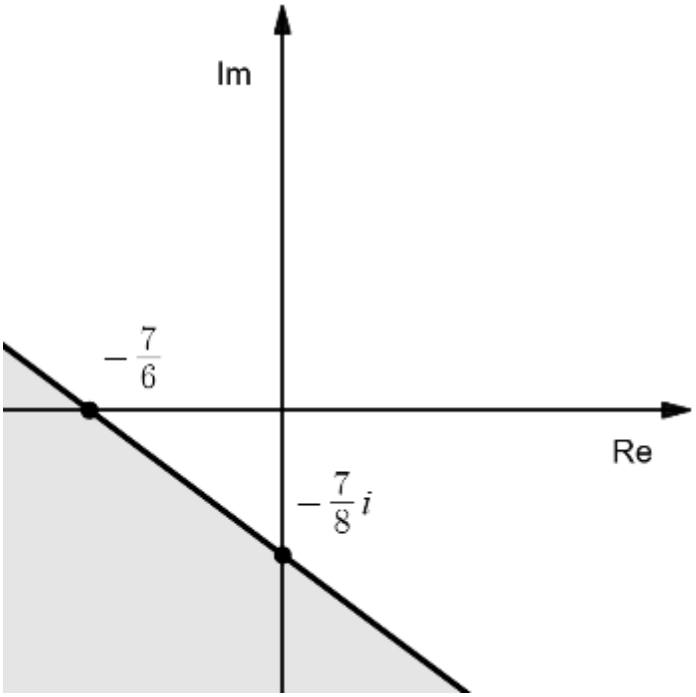
3 a)

Midpoint: $(7, 2)$



3 b)

Midpoint: $\left(\frac{3}{2}, -2\right)$ and Gradient: $\frac{4}{3}$

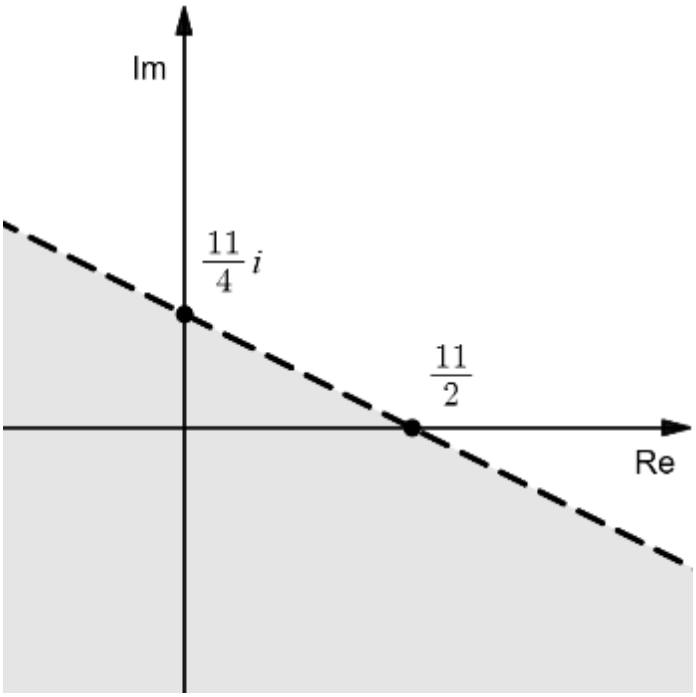


Line: $y + 2 = -\frac{3}{4}\left(x - \frac{3}{2}\right) \Rightarrow y = -\frac{3}{4}x - \frac{7}{8}$

When $x = 0$, $y = -\frac{7}{8}$ and when $y = 0$, $x = -\frac{7}{6}$

3 c)

Midpoint: $\left(-\frac{1}{2}, 3\right)$ and Gradient: 2

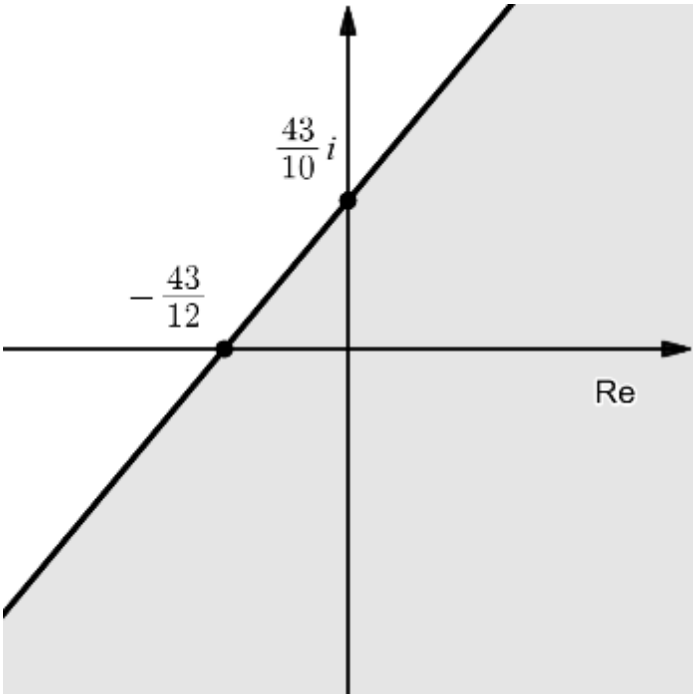


Line: $y - 3 = -\frac{1}{2}\left(x + \frac{1}{2}\right) \Rightarrow y = -\frac{1}{2}x + \frac{11}{4}$

When $x = 0$, $y = \frac{11}{4}$ and when $y = 0$, $x = \frac{11}{2}$

3 d)

Midpoint: $\left(1, \frac{11}{2}\right)$ and Gradient: $-\frac{5}{6}$



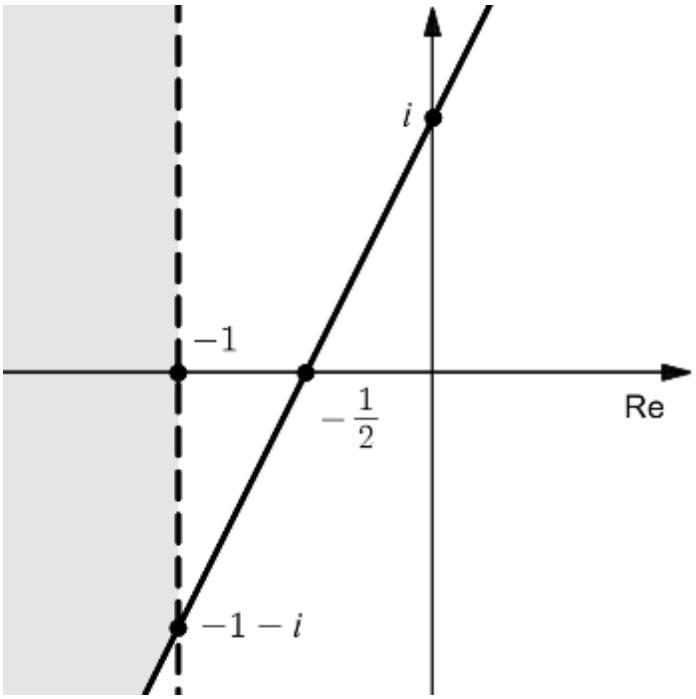
Line: $y - \frac{11}{2} = \frac{6}{5}(x - 1) \Rightarrow y = \frac{6}{5}x + \frac{43}{10}$

When $x = 0$, $y = \frac{43}{10}$ and when $y = 0$, $x = -\frac{43}{12}$

3 e)

Midpoint: $(-1, 0)$ gives $x = -1$

Midpoint: $(1, 3)$ and Gradient: $-\frac{1}{2}$



Line: $y-3=2(x-1) \Rightarrow y=2x+1$

When $x=0$, $y=1$ and when $y=0$, $x=-\frac{1}{2}$

4 a)

$$|z-2-2i|=2$$

$\tan\left(\frac{1}{1}\right)=\frac{\pi}{4}$ which gives $\arg(z-2-2i)=\frac{\pi}{4}$

$\arg(z-2-2i) \leq \frac{\pi}{4}$ and $|z-2-2i| \leq 2$

$$\left\{z \in \mathbb{C} : \arg(z-2-2i) \leq \frac{\pi}{4}\right\} \cap \{z \in \mathbb{C} : |z-2-2i| \leq 2\}$$

4 b)

$$|z-1|=|z-3|$$

$$|z-2|=1$$

$|z-2|<1$ and $|z-1| \leq |z-3|$

$$\{z \in \mathbb{C} : |z-2|<1\} \cap \{z \in \mathbb{C} : |z-1| \leq |z-3|\}$$

4 c)

$$|z-2i|=|z-2|$$

$\tan\left(\frac{1}{1}\right)=\frac{\pi}{4}$ which gives $\arg(z-2)=\frac{\pi}{4}$ and $\arg(z-2)=\frac{3\pi}{4}$

$|z-2i|>|z-2|$ and $\frac{\pi}{4} \leq \arg(z-2)<\frac{3\pi}{4}$

$$\{z \in \mathbb{C} : |z-2i|>|z-2|\} \cap \left\{z \in \mathbb{C} : \frac{\pi}{4} \leq \arg(z-2)<\frac{3\pi}{4}\right\}$$

4 d)

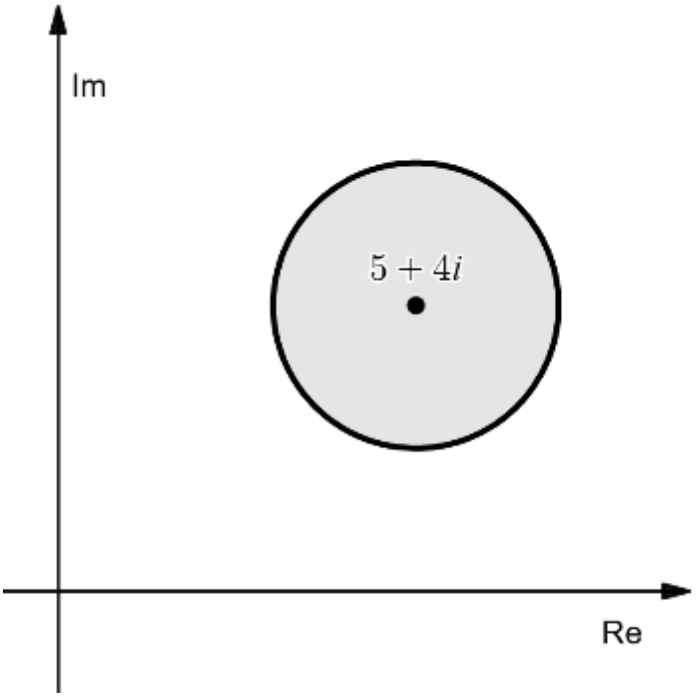
$$|z-1-i|=2$$

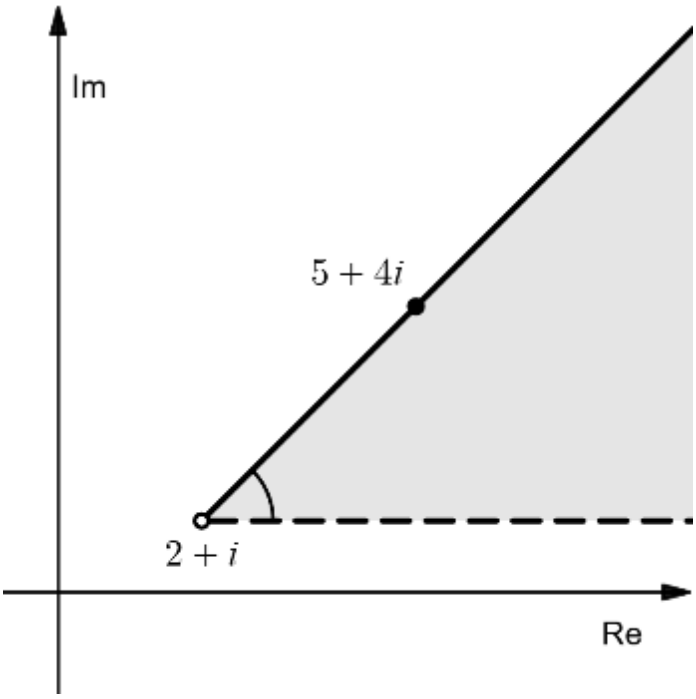
$$|z-1-i|=1$$

$|z-1-i| \leq 2$ and $|z-1-i|>1$

$$\{z \in \mathbb{C} : 1<|z-1-i| \leq 2\}$$

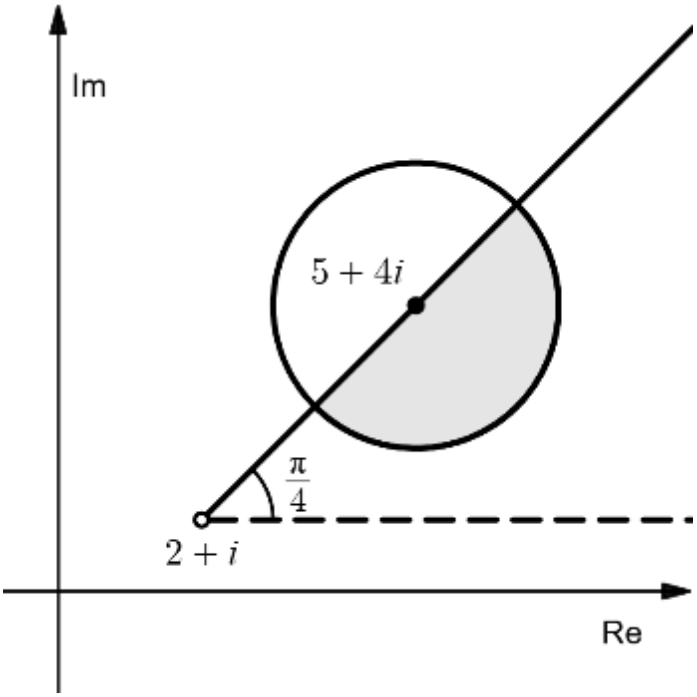
5 a)



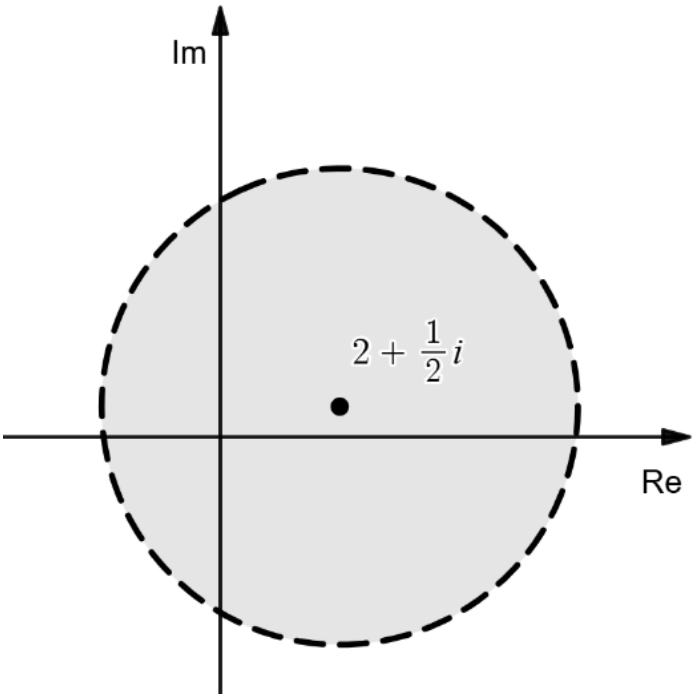


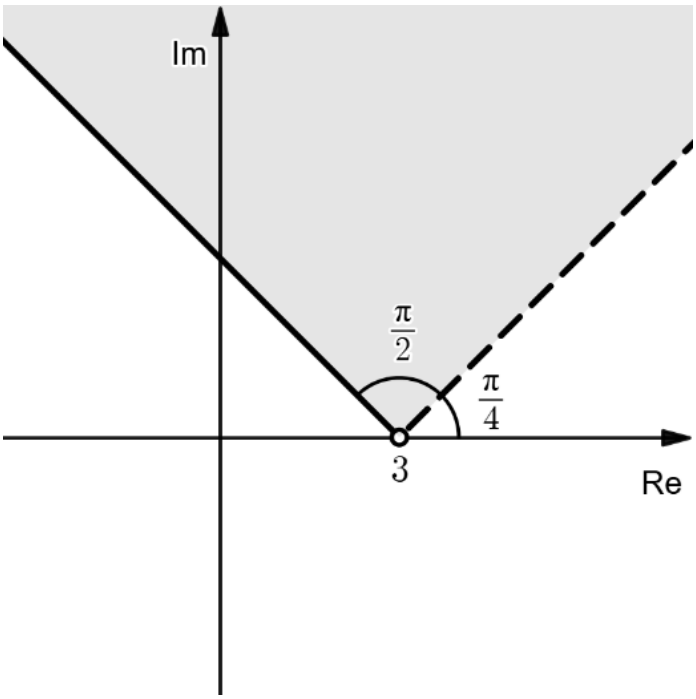
5 b)

$y - 1 = \tan \frac{\pi}{4}(x - 2) \Rightarrow y = x - 1$ so the line passes through $(5, 4)$



6 a)





6 b)

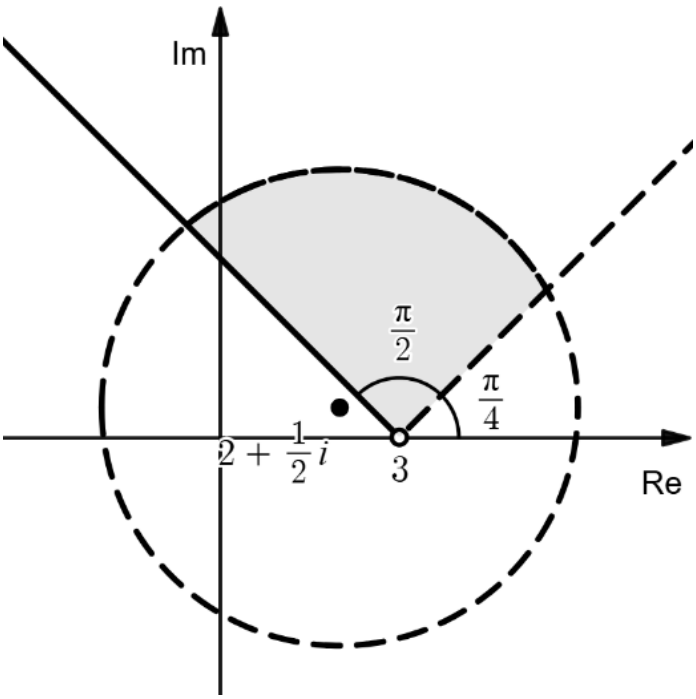
Consider the $\arg(z-3) = \frac{3\pi}{4}$ in cartesian form:

$$y = \tan\left(\frac{3\pi}{4}\right)(x-3) \Rightarrow y = -(x-3) = -x+3$$

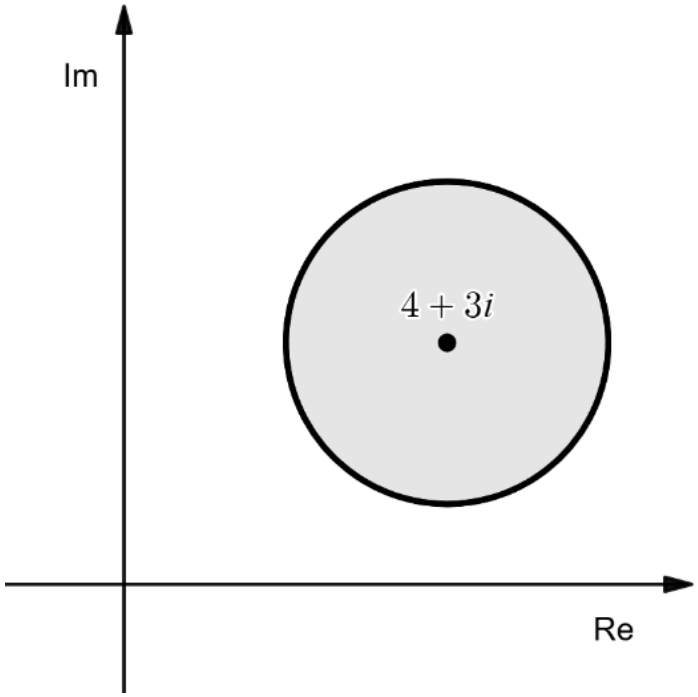
When $x = 2$, $y = -2+3 = 1$ so the line lies above the point $2 + \frac{1}{2}i$

So $2 + \frac{1}{2}i$ is not in the region.

6 c)



7 a)



7 b)

Find where $y = mx$ meets the equation of the circle $(x+4)^2 + (y+4)^2 = 4$

$$(x-4)^2 + (mx-3)^2 = 4$$
$$x^2 - 8x + 16 + m^2x^2 - 6mx + 9 - 4 = 0$$
$$(1+m^2)x^2 + (-8-6m)x + 21 = 0$$

Consider $b^2 - 4ac = 0$:

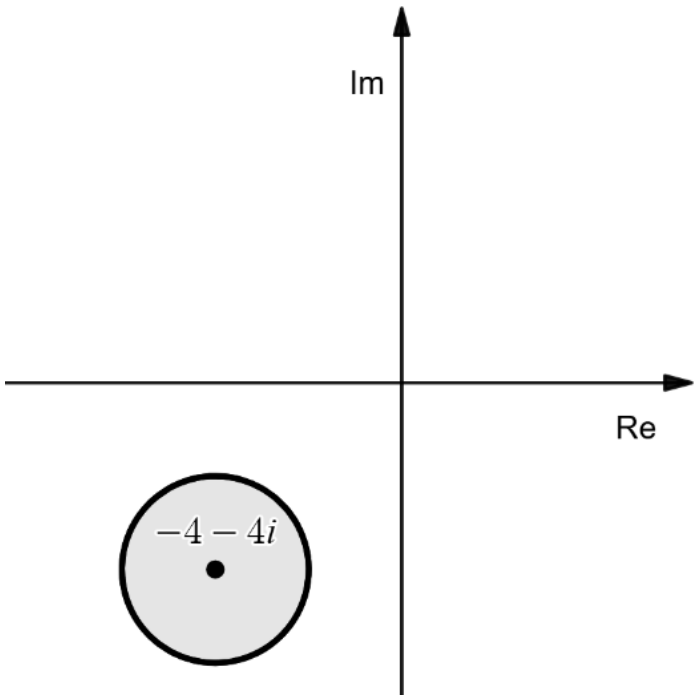
$$(-8-6m)^2 - 4 \times 21(1+m^2) = 0$$
$$36m^2 + 96m + 64 - 84 - 84m^2 = 0$$
$$-48m^2 + 96m - 20 = 0$$
$$12m^2 - 24m + 5 = 0$$
$$m = \frac{24 \pm \sqrt{576-240}}{24} = 1 \pm \frac{\sqrt{21}}{6}$$

$m = \tan\theta \therefore \arctan\left(1 + \frac{\sqrt{21}}{6}\right) = 1.055017\ldots$ and $\arctan\left(1 - \frac{\sqrt{21}}{6}\right) = 0.231984\ldots$

Therefore the least and greatest value of $\arg z$ are:

0.232 (3 SF) and 1.06 (3 SF)

8 a)



8 b)

Find where $y = mx$ meets the equation of the circle $(x+4)^2 + (y+4)^2 = 4$

$$(x+4)^2 + (mx+4)^2 = 4$$
$$x^2 + 8x + 16 + m^2x^2 + 8mx + 16 - 4 = 0$$
$$(1+m^2)x^2 + (8+8m)x + 28 = 0$$

Consider $b^2 - 4ac = 0$:

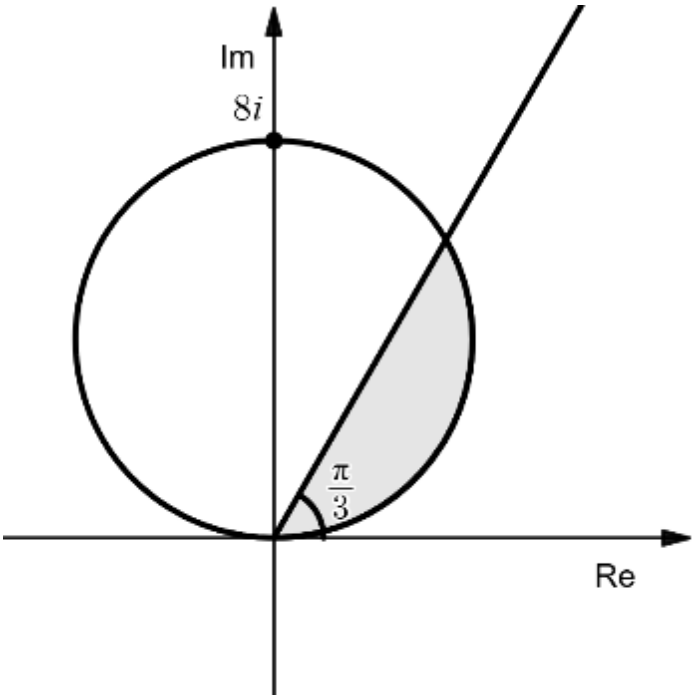
$$(8+8m)^2 - 4 \times 28(1+m^2) = 0$$
$$64m^2 + 128m + 64 - 112 - 112m^2 = 0$$
$$3m^2 - 8m + 3 = 0$$
$$m = \frac{8 \pm \sqrt{64-36}}{6} = \frac{4}{3} \pm \frac{\sqrt{7}}{3}$$

$m = \tan\theta \therefore \arctan\left(\frac{4}{3} + \frac{\sqrt{7}}{3}\right) = 1.146765\ldots$ and $\arctan\left(\frac{4}{3} - \frac{\sqrt{7}}{3}\right) = 0.424031\ldots$

Therefore the least and greatest value of $\arg z$ are:

$1.146765\ldots - \pi = -1.99$ (3 SF) and $0.424031\ldots - \pi = -2.72$ (3 SF)

9 a)

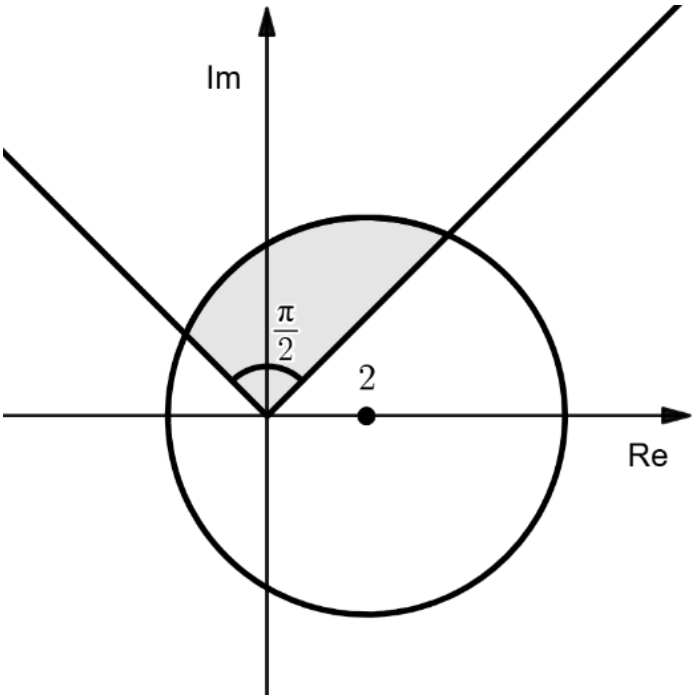


Maximum value of $|z|$ is the point where the circle and the line intersect:
$x^2 + (y-4)^2 = 16$ and $y = \tan\left(\frac{\pi}{3}\right) x = \sqrt{3}x$

$$x^2 + (\sqrt{3}x - 4)^2 = 16$$
$$x^2 + 3x^2 - 8\sqrt{3}x + 16 = 16$$
$$4x^2 - 8\sqrt{3}x = 0$$
$$4x(x - 2\sqrt{3}) = 0 \therefore x = 2\sqrt{3},\ y = 6$$

Maximum value of $|z| = \sqrt{(2\sqrt{3})^2 + 6^2} = 4\sqrt{3}$

9 b)



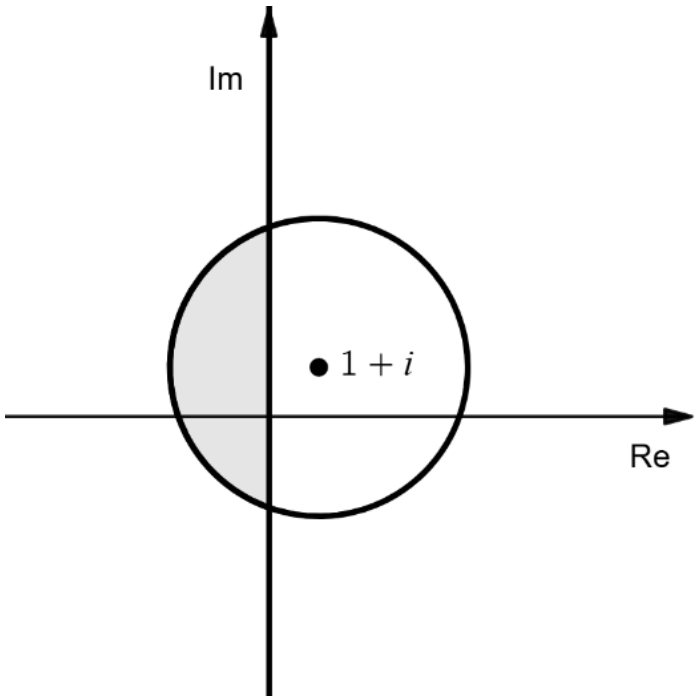
Maximum value of $|z|$ is the point where the circle and the line created by $\arg z = \frac{\pi}{4}$ intersect:
$(x-2)^2 + y^2 = 16$ and $y = \tan\left(\frac{\pi}{2}\right) x = x$

$$(x-2)^2 + x^2 = 16$$
$$x^2 - 4x + 4 + x^2 = 16$$
$$2x^2 - 4x - 12 = 0 \therefore x = 1 \pm \sqrt{7},\ y = 1 \pm \sqrt{7}$$

$$\text{Maximum value of } |z| = \sqrt{(1+\sqrt{7})^2 + (1+\sqrt{7})^2}$$
$$= \sqrt{16 + 4\sqrt{7}} = \sqrt{4(4+\sqrt{7})} = 2\sqrt{4+\sqrt{7}}$$

9 c)

Midpoint $(0, 1)$ so line is $x = 0$

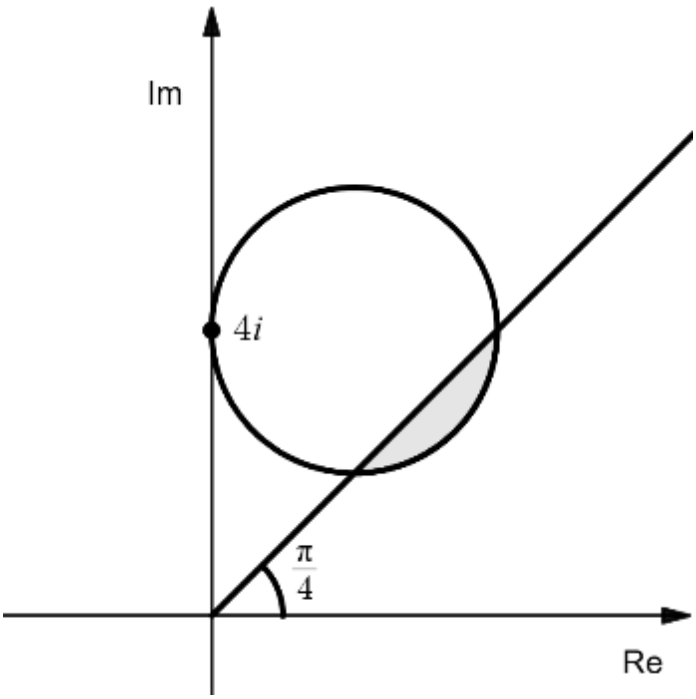


Maximum value of $|z|$ is the point where the circle crosses the imaginary axis:
$(x-1)^2 + (y-1)^2 = 9$ and $x = 0$

$$(y-1)^2 = 8$$
$$\therefore y = 1 \pm 2\sqrt{2},\ x = 0$$

Maximum value of $|z| = \sqrt{0^2 + (1+2\sqrt{2})^2} = 1 + 2\sqrt{2}$

10 a)

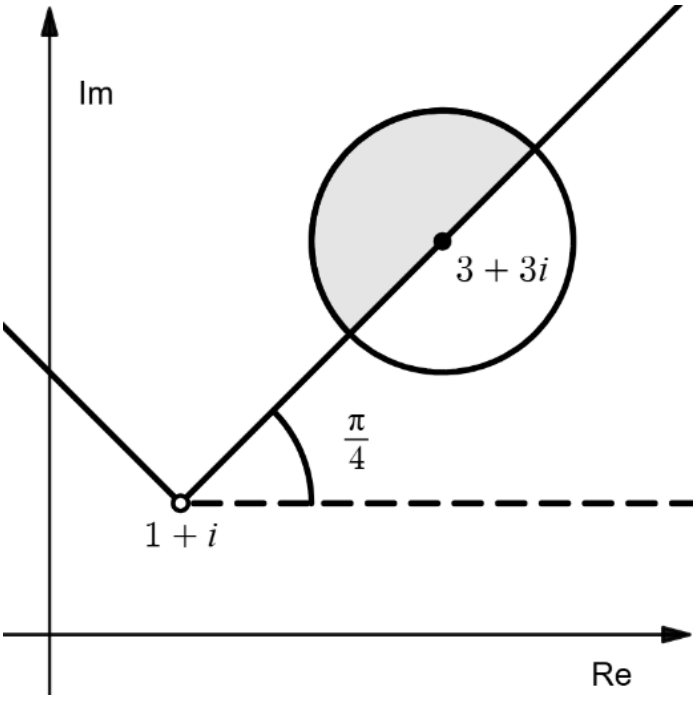


Minimum value of $|z|$ is the first point where the circle and the line intersect:
$(x-2)^2+(y-4)^2=4$ and $y=\tan\left(\frac{\pi}{4}\right)x=x$

$$(x-2)^2+(x-4)^2=4$$
$$x^2-4x+4+x^2-8x+16=4$$
$$2x^2-12x+16=0$$
$$\therefore x=2 \text{ or } x=4, \text{ so } (2,2) \text{ and } (4,4)$$

Minimum value of $|z|=\sqrt{2^2+2^2}=\sqrt{8}=2\sqrt{2}$

10 b)



Minimum value of $|z|$ is the first point where the $\arg z = \frac{\pi}{4}$ meets the circle:
$(x-3)^2 + (y-3)^2 = 2$ and $y = \tan\left(\frac{\pi}{4}\right) x = x$

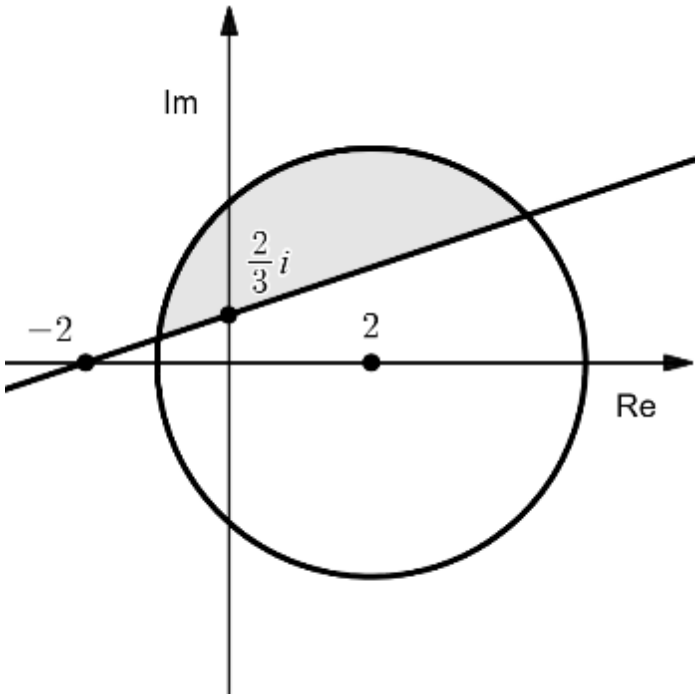
$$(x-3)^2 + (x-3)^2 = 2 \Rightarrow 2(x-3)^2 = 2 \Rightarrow (x-3)^2 = 1$$

$\therefore x = 2$ or $x = 4$, so $(2, 2)$ and $(4, 4)$
Minimum value of $|z| = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$

10 c)

Midpoint: $(-5, -1)$ Gradient: -3

$$y + 1 = \frac{1}{3}(x+5) \Rightarrow y = \frac{1}{3}x + \frac{2}{3}$$

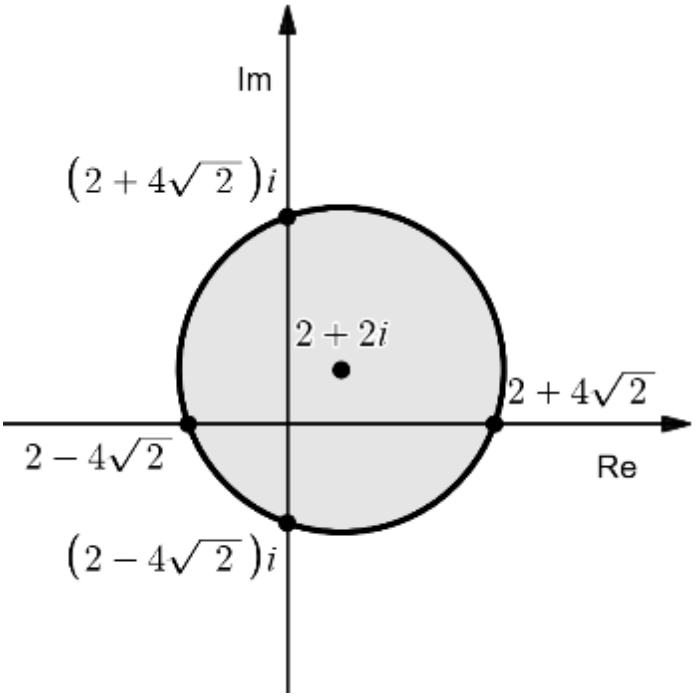


Minimum value of $|z|$ is the shortest distance to the line:
Find the perpendicular line passing through $(0, 0)$: $y = -3x$
Find the point where these two lines intersect: $y = \frac{1}{3}x + \frac{2}{3}$ and $y = -3x$
$\frac{1}{3}x + \frac{2}{3} = -3x \therefore x = -\frac{1}{5}$, $y = \frac{3}{5}$ so minimum value of $|z| = \sqrt{\left(-\frac{1}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \frac{\sqrt{10}}{5}$

11 a)

$$(x-2)^2 + (y-2)^2 = 36$$

When $x = 0$, $(y-1)^2 = 32 \Rightarrow y = 2 \pm 4\sqrt{2}$
When $y = 0$, $(x-2)^2 = 35 \Rightarrow x = 2 \pm 4\sqrt{2}$



11 b)

$$\left|9\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)\right|=9$$

Distance to the maximum $|z|$ with angle $\frac{\pi}{4}$ is $\sqrt{2^2+2^2}+6=6+2\sqrt{2}$

Since $6+2\sqrt{2}<9$ then $9\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$ is outside the region.

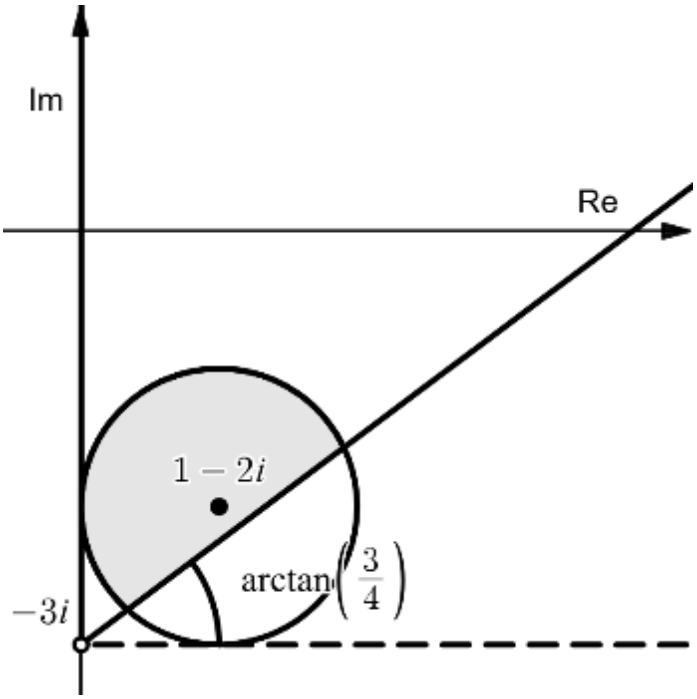
11 c)

$$\left|4\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)\right|=4$$

Distance away from when $\mathrm{Re}(z)=0$ is $\left|4\cos\frac{3\pi}{4}\right|=2\sqrt{2}$

Since $|2\sqrt{2}|<|2-4\sqrt{2}|$ then $4\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$ lies inside the region.

12 a)



12 b)

Minimum value of $|z|$ is the shortest distance to the circle along the line which passes through the origin and $1-2i$:

$(x-1)^2+(y+2)^2=1$ and $y=-2x$

$$(x-1)^2+(-2x+2)^2=1$$
$$x^2-2x+1+4x^2-8x+4=1$$
$$5x^2-10x+4=0$$
$$\therefore x=1\pm\frac{\sqrt{5}}{5},\ y=-2\mp\frac{2\sqrt{5}}{5}$$

$$\text{so minimum value of } |z|=\sqrt{\left(1-\frac{\sqrt{5}}{5}\right)^2+\left(-2+\frac{2\sqrt{5}}{5}\right)^2}$$
$$=\sqrt{6-2\sqrt{5}}=\sqrt{5}-1$$
$$\approx 1.23 \text{ (3 SF)}$$

13 a)

The circle $x^2+y^2=25$ and the line $y=\tan\left(\frac{\pi}{3}\right)x=\sqrt{3}x$ when:

$$x^2+(\sqrt{3}x)^2=25$$
$$x^2+3x^2=25$$
$$x^2=\frac{25}{4}$$

$x=\pm\frac{5}{2}$ so $x=\frac{5}{2}$ and $y=\frac{5\sqrt{3}}{2}$

The circle $x^2 + y^2 = 25$ and the line $y = \tan\left(\frac{3\pi}{4}\right)x = -x$ when:

$$x^2 + (-x)^2 = 25$$
$$2x^2 = 25$$
$$x^2 = \frac{25}{2}$$

$x = \pm\sqrt{\frac{25}{2}}$ so $x = -\frac{5\sqrt{2}}{2}$ and $y = \frac{5\sqrt{2}}{2}$

So A is $-\frac{5\sqrt{2}}{2} + \frac{5\sqrt{2}}{2}i$ and B is $\frac{5}{2} + \frac{5\sqrt{3}}{2}i$

13 b)

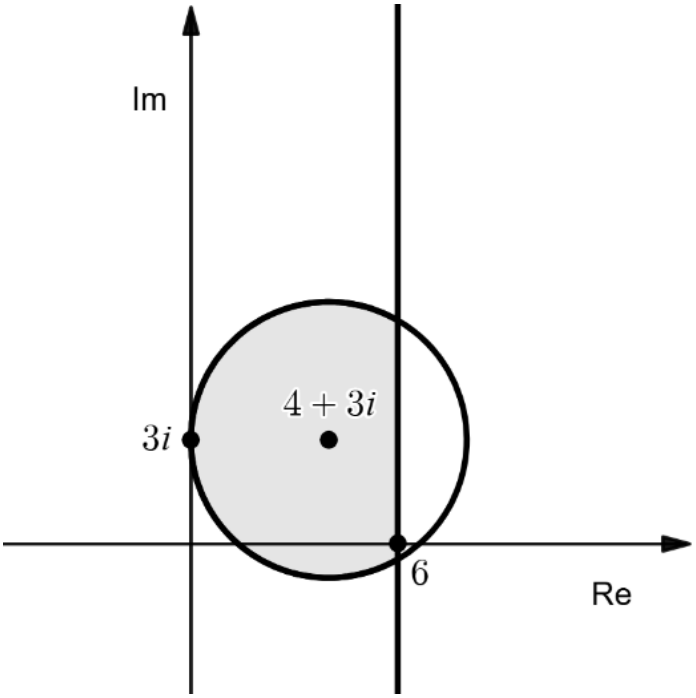
Area of sector $= \frac{1}{2} \times 5^2 \times \left(\frac{3\pi}{4} - \frac{\pi}{3}\right) = \frac{125\pi}{24}$ units^2

Area of circle $= 5^2\pi = 25\pi$ units^2

Shaded area $= 25\pi - \frac{125\pi}{24} = \frac{475\pi}{24}$ units^2

14 a)

Midpoint: $(6, 0)$

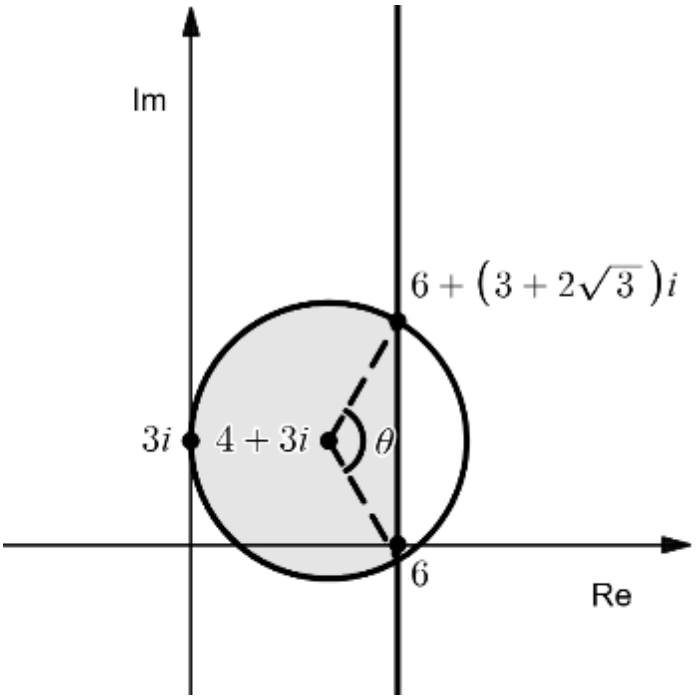


14 b)

Radius of circle: 4

Cartesian equation of circle: $(x-4)^2 + (y-3)^2 = 16$

So when $x = 6$, $y = 3 \pm 2\sqrt{3}$



Let $\phi = \frac{1}{2}\theta$

$$\tan\phi = \frac{3+2\sqrt{3}-3}{6-4} = \sqrt{3}$$

Angle: $\theta = 2 \times \arctan\sqrt{3} = 2\arctan(\sqrt{3}) = \frac{2\pi}{3}$

Area is Circle – Segment

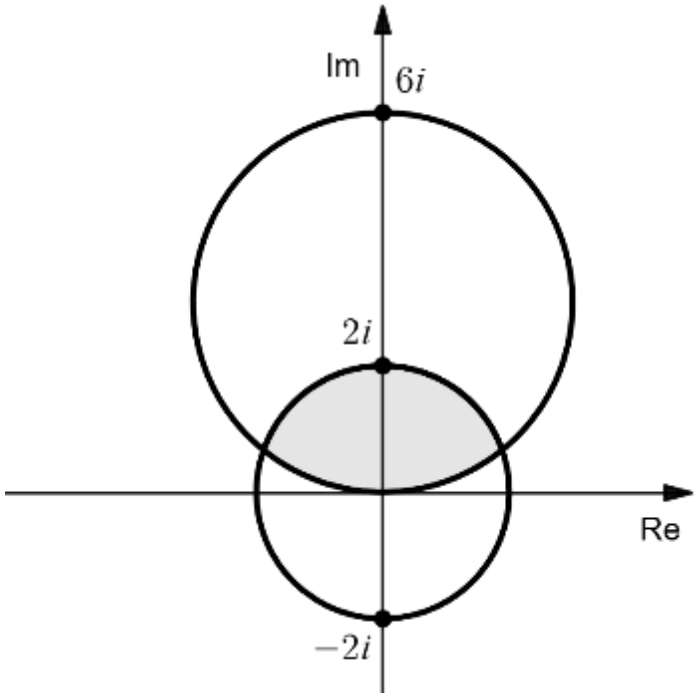
Area of segment:

$$\frac{1}{2}r^2(\theta - \sin\theta) = \frac{1}{2} \times 4^2\left(\frac{2\pi}{3} - \sin\frac{2\pi}{3}\right) = \frac{16\pi}{3} - 4\sqrt{3}$$

Area of circle: $\pi r^2 = 16\pi$

Area of shaded region: $16\pi - \frac{16\pi}{3} + 4\sqrt{3} = \frac{32\pi}{3} + 4\sqrt{3}$ as req.

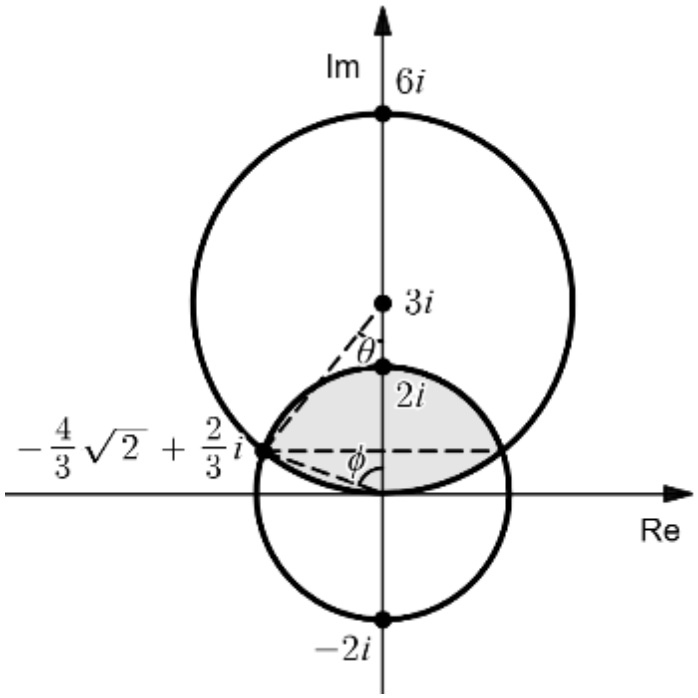
15)



Points of intersection are where $x^2 + (y-3)^2 = 9$ and $x^2 + y^2 = 4$ intersect:

$$4 - y^2 + (y-3)^2 = 9$$
$$4 - y^2 + y^2 - 6y + 9 = 9$$

$\therefore y = \frac{2}{3}$ so $\left(-\frac{4\sqrt{2}}{3}, \frac{2}{3}\right)$ and $\left(\frac{4\sqrt{2}}{3}, \frac{2}{3}\right)$



$\tan\theta = \frac{\frac{4\sqrt{2}}{3}}{3-\frac{2}{3}} = \frac{4\sqrt{2}}{7}$ $\therefore$ the required angle is $2\arctan\left(\frac{4\sqrt{2}}{7}\right)$

$\therefore$ segment of larger circle is

$$\frac{1}{2} \times 3^2 \times 2\arctan\left(\frac{4\sqrt{2}}{7}\right) - \frac{1}{2} \times 3^2 \sin\left(2\arctan\left(\frac{4\sqrt{2}}{7}\right)\right) = 1.7172888\ldots$$

$\tan\phi = \frac{\frac{4\sqrt{2}}{3}}{\frac{2}{3}} = 2\sqrt{2}$ $\therefore$ the required angle is $2\arctan(2\sqrt{2})$

$\therefore$ segment of smaller circle is

$$\frac{1}{2} \times 2^2 \times 2\arctan(2\sqrt{2}) - \frac{1}{2} \times 2^2 \sin(2\arctan(2\sqrt{2})) = 0.5763007515\ldots$$

Total is area is ≈ 2.29 units^2 (3 SF)